Introduction to Probability and Applied Statistics

Week 9: Linear Regression II

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Roadmap

Notation and where we are

The Gauss–Markov assumptions

Omitted variable bias

Three routes to a zero conditional mean

Wrapping up

Notation: lines and coefficients

Symbol	What it is
β_0, β_1	The population intercept and slope.
$Y = \beta_0 + \beta_1 X + \varepsilon$	The population regression function, with error ε.
$\hat{\beta}_0, \hat{\beta}_1$	The OLS estimates from one sample. A new sample gives new values.
$Y = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\varepsilon}$	The sample regression function, with residual $\hat{\varepsilon}$.
$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$	The fitted value for unit i.
$\hat{\varepsilon}_i = Y_i - \hat{Y}_i$	The residual for unit i.

Gailmard writes the estimated line as $a + bx$: his a is $\hat{\beta}_0$ and his b is $\hat{\beta}_1$.

The OLS formulas

BIVARIATE OLS

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X, Y)}{s_X^2} = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

$\hat{\beta}_0$ and $\hat{\beta}_1$ are computed from a sample, so both are estimators.

The Gauss–Markov theorem, informally

GAUSS–MARKOV THEOREM

Under a set of assumptions, the ordinary least squares estimator is the **best linear unbiased estimator** of the population slope and intercept.

"OLS is **BLUE**."

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The set-up

- **Variables.** An explanatory variable X and an outcome Y. With more than one explanatory variable, $X_1, X_2, \ldots, X_k$.
- **Parameters.** The population intercept β_0 and slope β_1.
- **Data.** A sample of n units, (X_i, Y_i) for $i = 1, \ldots, n$. From it we compute $\hat{\beta}_0$ and $\hat{\beta}_1$.

A MODEL FOR THE DATA

From here on, we assume that the data were produced by

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

We assume this equation describes the true relationship between X and Y in the world.

Error and residual

ERROR

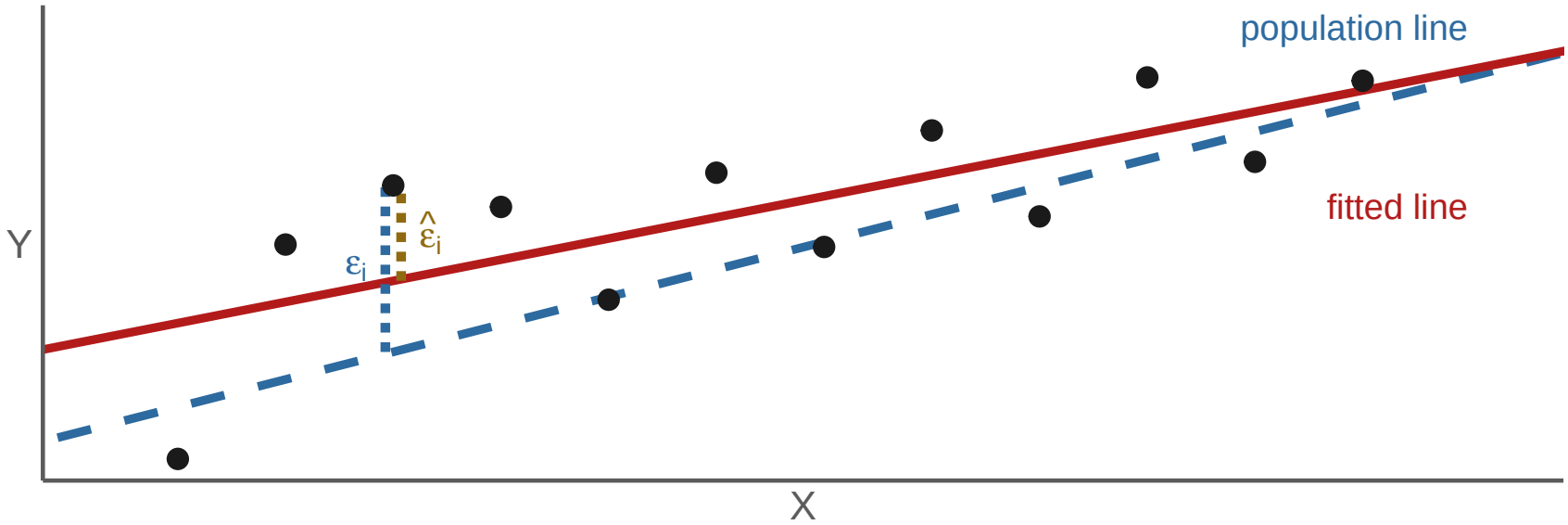
$$\varepsilon_i = Y_i - \beta_0 - \beta_1 X_i$$

The distance from Y_i to the **population** line. ε_i is the part of Y_i that the population line does not account for. We never observe ε_i.

RESIDUAL

$$\hat{\varepsilon}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$$

The distance from Y_i to the fitted line (fitted to the **sample**). We compute $\hat{\varepsilon}_i$ from the data.



Assumption 1: linearity

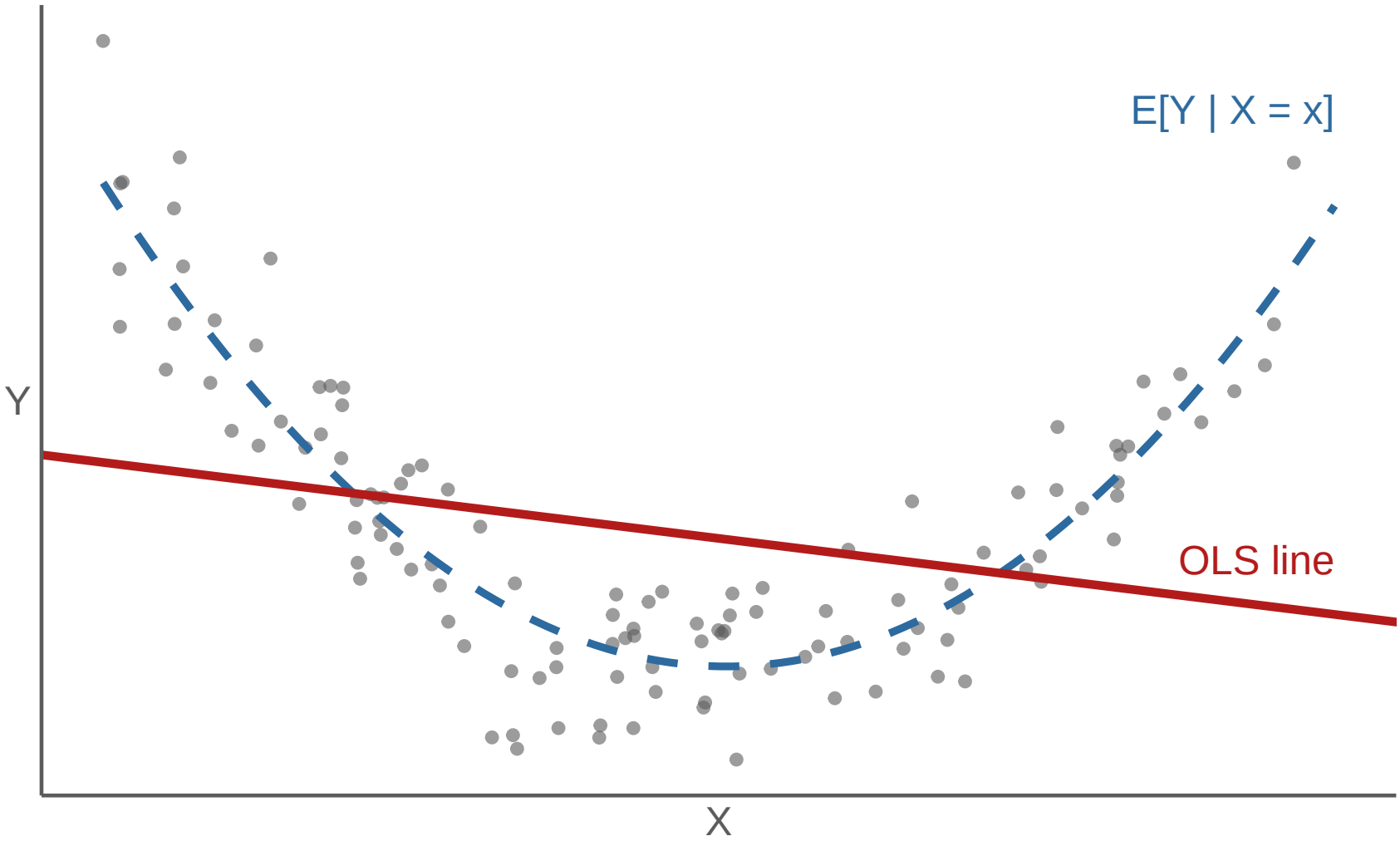
WHAT IT SAYS $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$

WHAT IT MEANS Y is a linear function of the parameters β_0 and β_1, plus an error. The model describes how the data were produced.

HOW IT FAILS The population relationship is curved, and we fit a straight line.

"Linear" refers to the βs. $Y_i = \beta_0 + \beta_1 X_i^2 + \varepsilon_i$ satisfies Assumption 1, with X_i^2 as the explanatory variable. $Y_i = \beta_0 + X_i^{\beta_1} + \varepsilon_i$ does not.

A curved relationship, a straight line



Assumption 2: random sampling

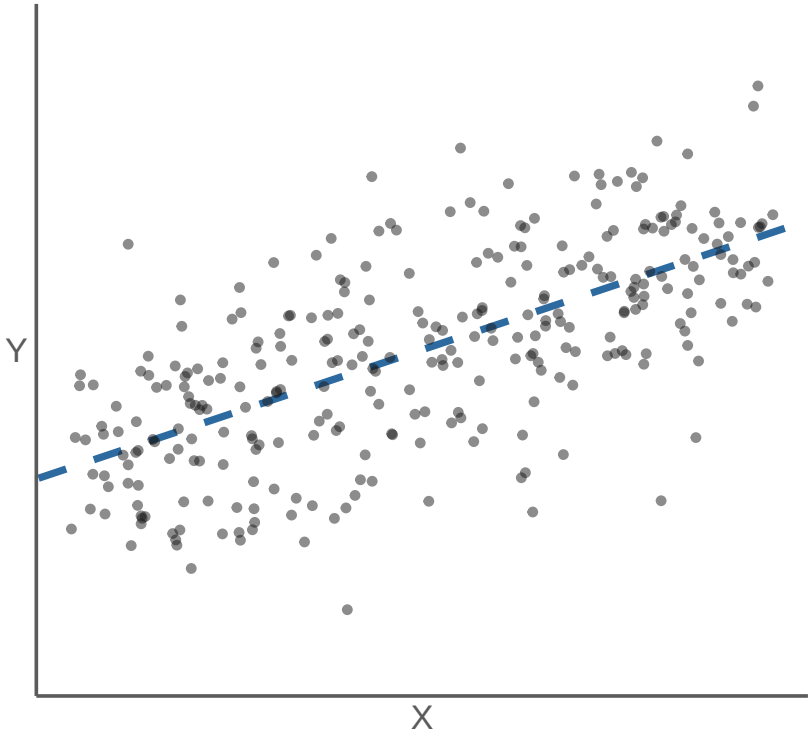
WHAT IT SAYS (X_i, Y_i) for $i = 1, \ldots, n$ is an iid sample of size n.

WHAT IT MEANS Each unit is drawn at random from the same population, independently of the others.

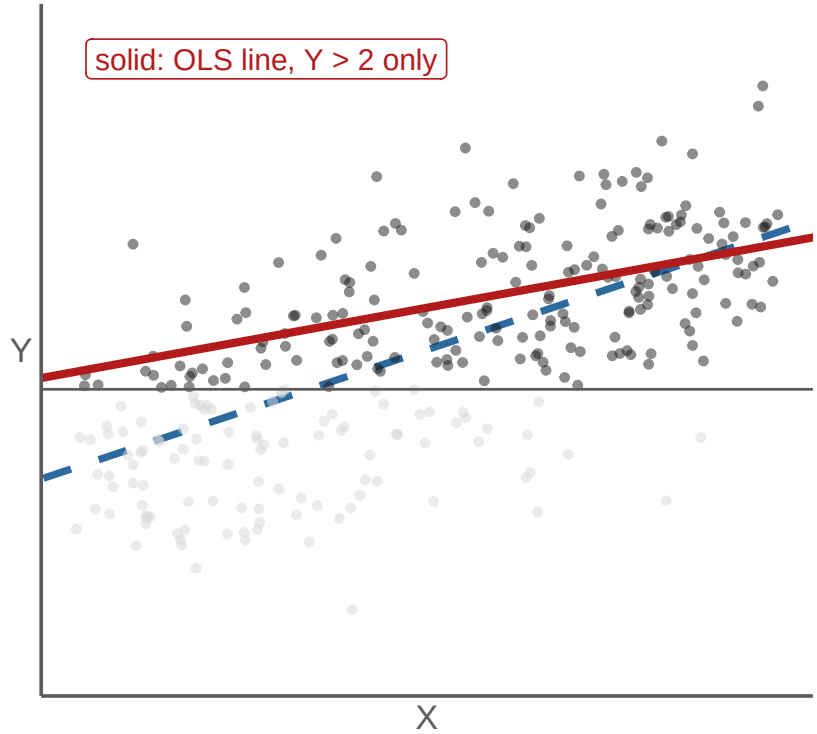
HOW IT FAILS Units depend on each other: one unit's (X_i, Y_i) is related to another unit's (X_j, Y_j). Or the sample is selected on Y (e.g., we only observe units with large values of Y).

Selecting the sample on *Y*

Random sample



Only units with Y > 2



Dashed: the population line.

Assumption 3: variation in X (no perfect collinearity)

WHAT IT SAYS $\sum_i (X_i - \bar{X})^2 > 0$. With several explanatory variables: no X is an exact linear function of the others.

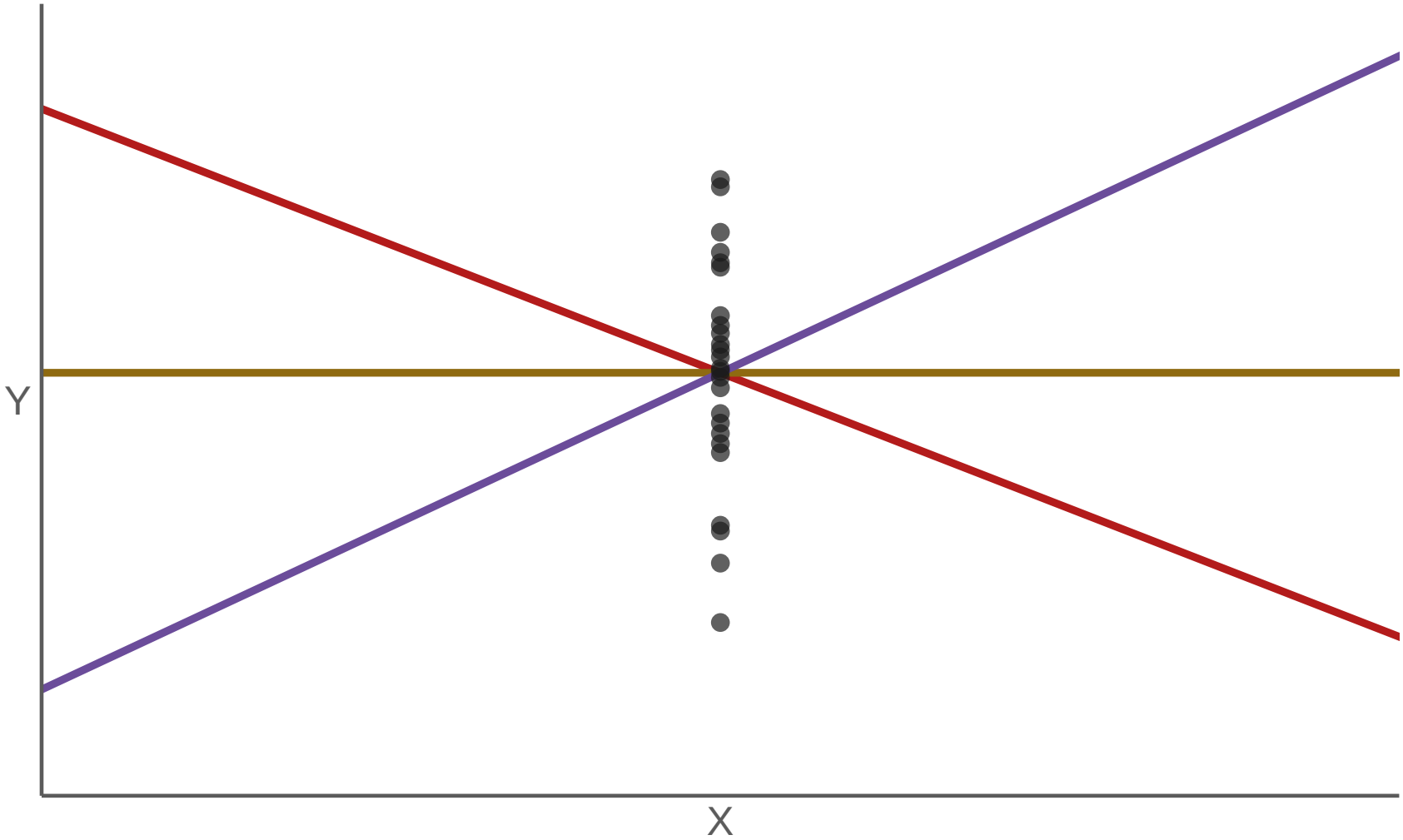
DEFINITION Explanatory variables are **perfectly collinear** when one is an exact linear function of the others, e.g. $X_2 = 3X_1$ or $X_2 = 5 + 2X_1$.

WHAT IT MEANS X takes at least two different values in the sample. With several Xs, each carries information the others do not.

HOW IT FAILS X is the same for every unit. Or, with several Xs, they are perfectly collinear.

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{0}{0} \quad \text{when every } X_i = \bar{X}.$$

Every X_i the same: which line?



Every line through $(\bar{X}, \bar{Y})$ has the same SSR. No line is the unique minimum.

Assumption 4: zero conditional mean

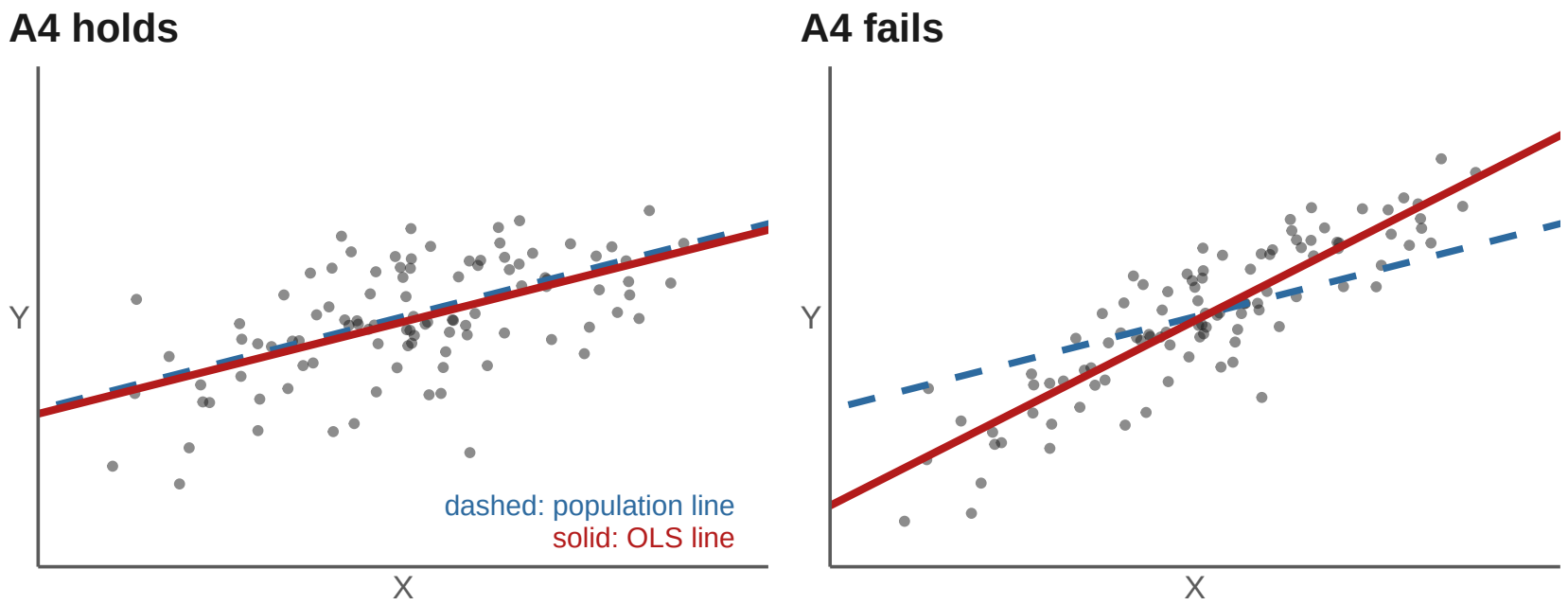
WHAT IT SAYS $\mathbb{E}[\varepsilon \mid X = x] = 0$ for every value x.

WHAT IT MEANS At every value of X, the error averages to zero. The average of the other influences on Y is the same at every value of X.

HOW IT FAILS Some other influence on Y is correlated with X. Or Y also affects X.

How Assumption 4 fails: an omitted influence

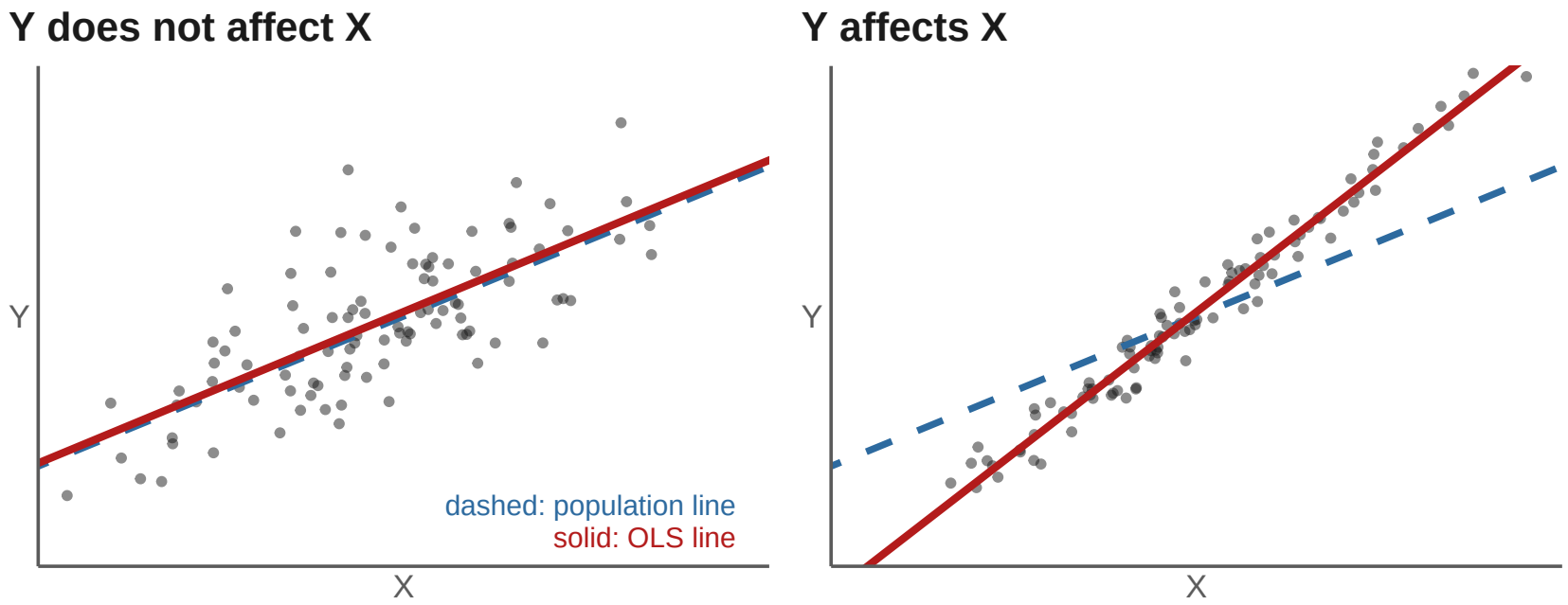
ε contains some variable U that affects Y and is correlated with X. Then the average of ε differs across values of X.



Both panels: $Y_i = 1 + 0.5X_i + \varepsilon_i$, $n = 100$. Right panel: ε_i contains a variable U_i with $\mathbb{E}[U \mid X = x] = 0.5x$.

How Assumption 4 fails: Y affects X

If $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ and X_i also depends on Y_i, then X_i depends on ε_i.



Both panels: $Y_i = 1 + 0.5X_i + \varepsilon_i$, $n = 100$. Right panel: $X_i = (Y_i - 1) + v_i$, where v_i is noise.

Two ways Assumption 4 fails

- **An omitted influence.** ε contains some variable U that affects Y and is correlated with X.
- Y **affects** X **(reverse causality).** X_i depends on Y_i, so X_i depends on ε_i.

Assumption 4 concerns ε, which we never observe. No test on the sample can confirm it.

By construction, OLS residuals average zero and have zero correlation with X. So the residuals cannot confirm that A4 holds.

OLS is unbiased

Suppose all four assumptions hold. What do we get?

UNBIASEDNESS OF OLS

If Assumptions 1–4 hold,

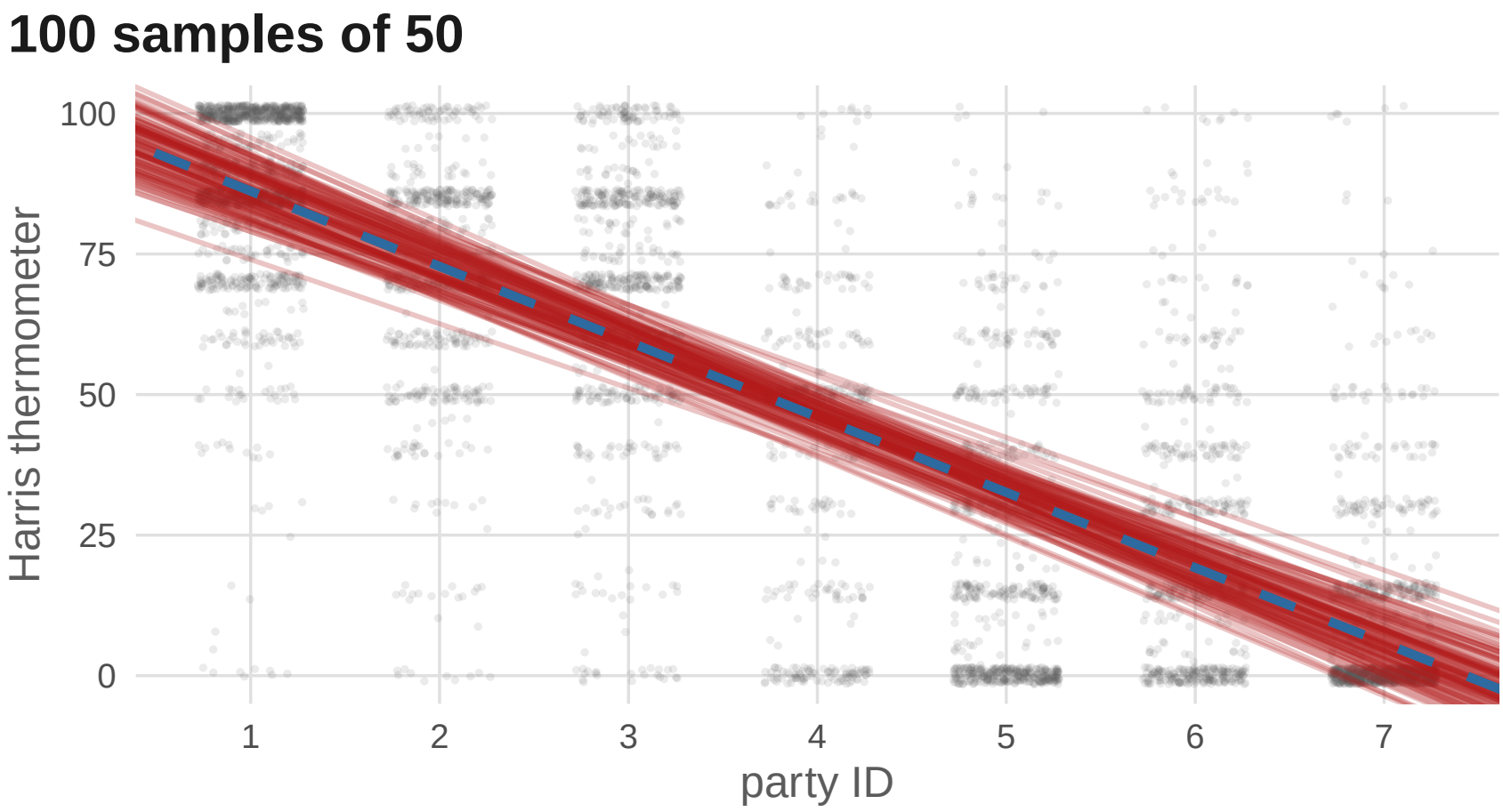
$$\mathbb{E}[\hat{\beta}_0] = \beta_0, \qquad \mathbb{E}[\hat{\beta}_1] = \beta_1.$$

- A3 is needed for $\hat{\beta}_1$ to exist. A3 concerns the $\boldsymbol{X}$ values in our sample, and we can check it.
- Given A3, A1, A2 and A4 together guarantee that the sampling distribution of $\hat{\beta}_1$ is centered on β_1.
- Unbiasedness does not depend on A5 or A6.

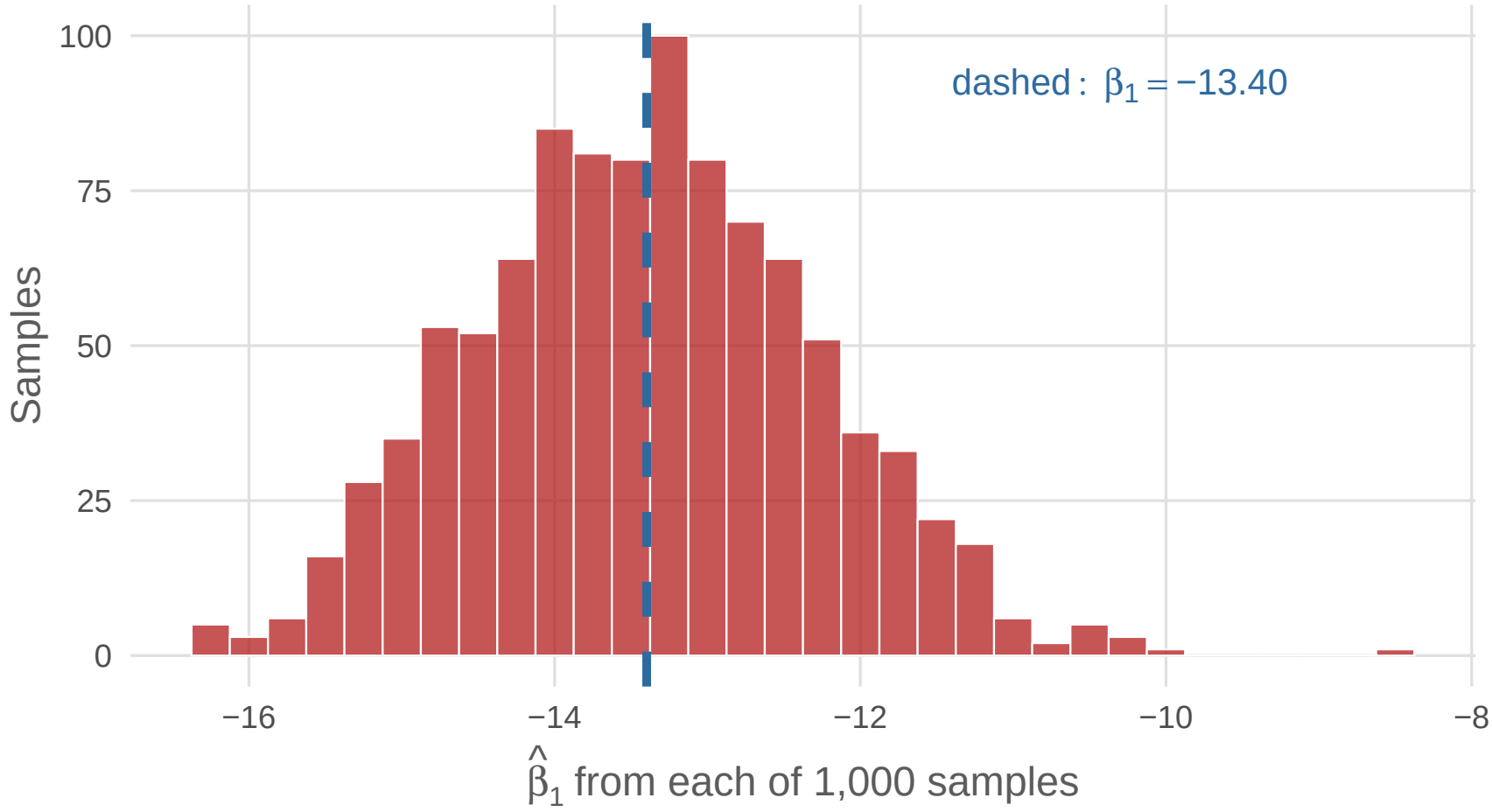
Next: a simulation, to see what $\mathbb{E}[\hat{\beta}_1] = \beta_1$ means.

Drawing many samples

Treat the 5,467 ANES respondents as the population. The population line: $\beta_0 = 99.59$, $\beta_1 = -13.40$. Draw a sample of 50, fit OLS, repeat.



The sampling distribution of $\hat{\beta}_1$



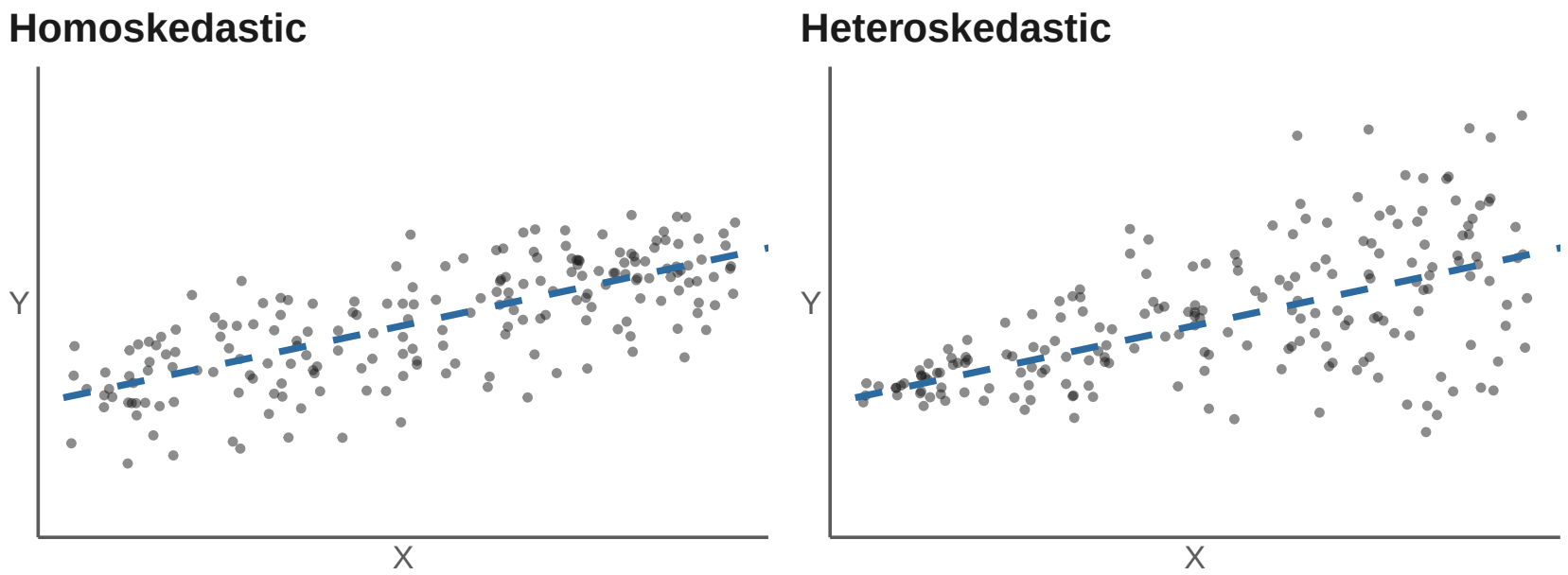
Mean of the 1,000 estimates: −13.40.

Assumption 5: homoskedasticity

WHAT IT SAYS $\mathbb{V}[\varepsilon \mid X = x] = \sigma_\varepsilon^2$ for every value x.

WHAT IT MEANS The errors have the same variance, σ_ε^2, at every value of X.

HOW IT FAILS The spread of Y around the population line is larger at some values of X than at others: **heteroskedasticity**.



The variance of $\hat{\beta}_1$

SAMPLING VARIANCE OF THE OLS SLOPE

If Assumptions 1–5 hold,

$$\mathbb{V}[\hat{\beta}_1 \mid X] = \frac{\sigma_\varepsilon^2}{\sum_i (X_i - \bar{X})^2}.$$

σ_ε^2: the variance of the errors.

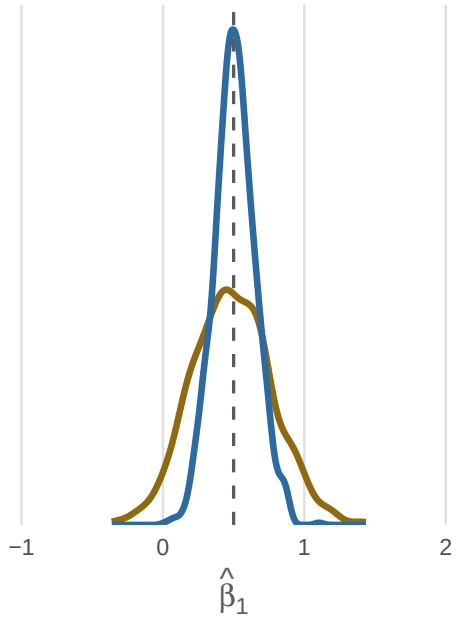
The variance is smaller when

- n is larger: the sum in the denominator has more terms;
- the X_i are more spread out: each $(X_i - \bar{X})^2$ is larger;
- σ_ε^2 is smaller: the points sit closer to the population line.

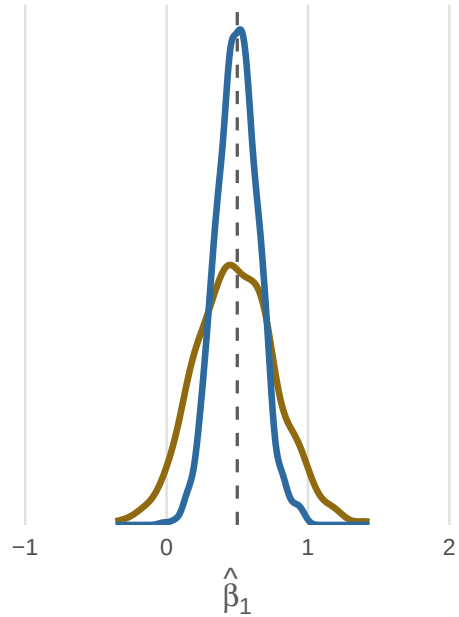
Three ways to a smaller variance

$$\mathbb{V}[\hat{\beta}_1 \mid X] = \frac{\sigma_\varepsilon^2}{\sum_i (X_i - \bar{X})^2}$$

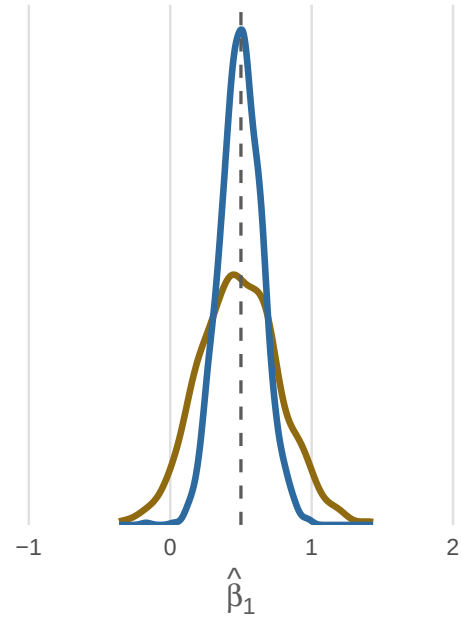
More units: $n = 200$



More spread: SD of $X = 2$



Smaller error: $\sigma_\varepsilon = 1$



Each curve: 1,000 slopes from $Y_i = 1 + 0.5X_i + \varepsilon_i$. Gold, in every panel: $n = 50$, SD of $X = 1$, $\sigma_\varepsilon = 2$. Blue: one change, named above the panel.

OLS is BLUE

GAUSS–MARKOV THEOREM

If Assumptions 1–5 hold, OLS is the **best linear unbiased estimator** of β_0 and β_1.

- **Linear.** $\hat{\beta}_1 = \sum_i w_i Y_i$, a weighted sum of the Y_i. The weights $w_i = (X_i - \bar{X}) / \sum_j (X_j - \bar{X})^2$ depend only on the Xs.
- **Unbiased.** $\mathbb{E}[\hat{\beta}_1] = \beta_1$.
- **Best.** Among all linear unbiased estimators of β_1, OLS has the smallest variance. In earlier terms: OLS is the most efficient linear unbiased estimator.

If Assumption 5 fails, OLS is still unbiased, but it is no longer "best". Some other linear unbiased estimator can then have a smaller variance.

Estimating σ_ε^2 and the standard error

σ_ε^2 is the variance of the errors, which we never observe. We use the residuals in their place.

ESTIMATES

$$\hat{\sigma}_\varepsilon^2 = \frac{\text{SSR}}{n-2} = \frac{\sum_i \hat{\varepsilon}_i^2}{n-2}, \qquad \widehat{\text{SE}}[\hat{\beta}_1] = \frac{\hat{\sigma}_\varepsilon}{\sqrt{\sum_i (X_i - \bar{X})^2}}.$$

$n-2$: two coefficients are estimated from the data, as s^2 divides by $n-1$ after estimating one mean.

$\hat{\sigma}_\varepsilon$ is the "Residual standard error" in `summary()`.

Assumption 6: normality of the errors

Under Assumptions 1–5, we know the mean of $\hat{\beta}_1$ and have a formula for its standard error. To build a confidence interval, we also need the distribution of $\hat{\beta}_1$ across samples.

WHAT IT SAYS $\varepsilon \mid X = x \sim N(0, \sigma_\varepsilon^2)$ for every value x.*

WHAT IT MEANS Given the Xs, the sampling distribution of $\hat{\beta}_1$ is exactly normal. The t statistic follows t_{n-2} exactly, at any n.

HOW IT FAILS Skewed or heavy-tailed errors at some value of X.

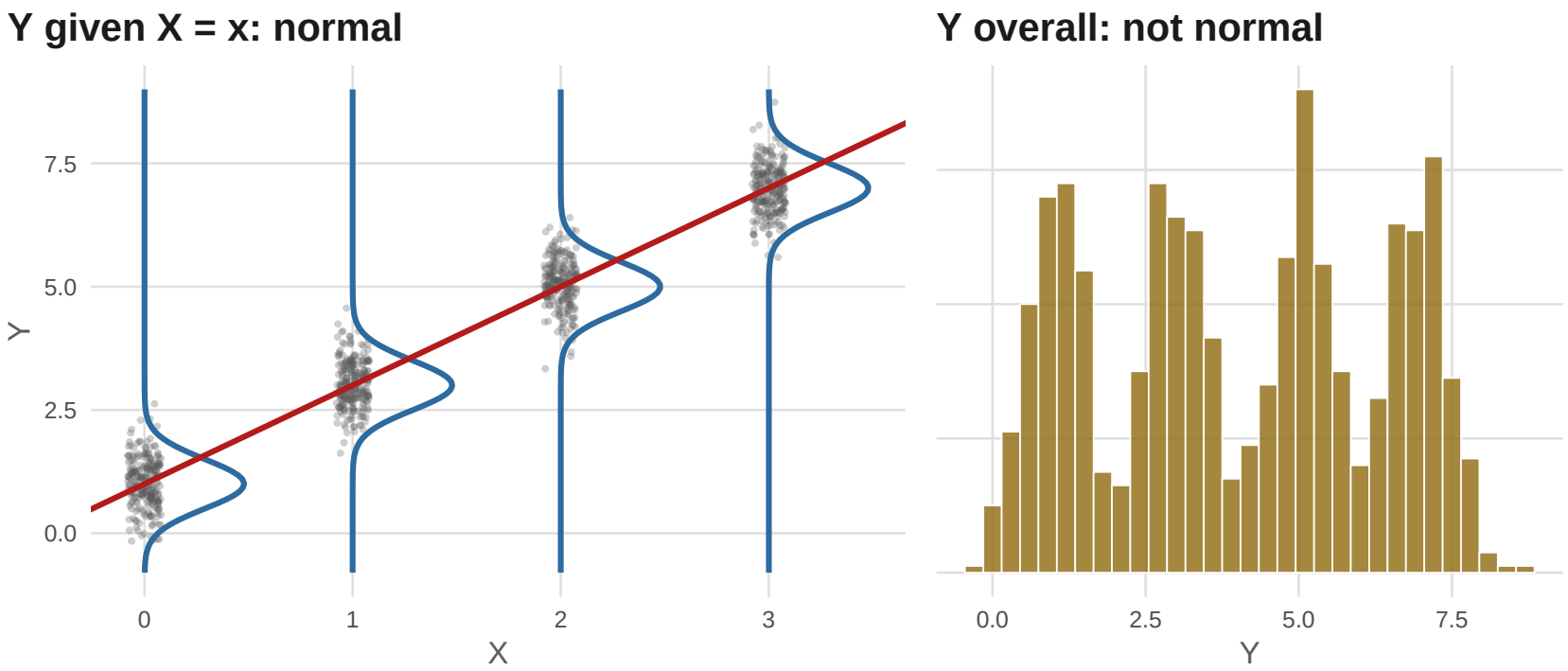
*If n is large, we do not need A6. By the central limit theorem, the sampling distribution of $\hat{\beta}_1$ is approximately normal under A1–A5.

With A1–A6, exactly (with A1–A5 and a large n, approximately),

$$t_{\text{obs}} = \frac{\hat{\beta}_1 - \beta_1}{\widehat{\text{SE}}[\hat{\beta}_1]} \sim t_{n-2}, \qquad \text{95\% CI: } \hat{\beta}_1 \pm t_{n-2,.025}\, \widehat{\text{SE}}[\hat{\beta}_1].$$

Normality is **not** a Gauss–Markov assumption. OLS is BLUE without it.

Normal at each x, not overall



When ***n*** is small, we need the errors to be normal at each value of ***X***. A histogram of ***Y*** alone cannot check that condition.

The assumptions we need for each claim

Claim	Assumptions
The OLS line exists	A3
OLS is unbiased	A1–A4
OLS is BLUE	A1–A5
t statistics and CIs, large *n*	A1–A5
t statistics and CIs, small *n*	A1–A6

A1 Linearity
A2 Random sampling
A3 Variation in X
A4 Zero conditional mean
A5 Homoskedasticity
A6 Normality of the errors (not a Gauss–Markov assumption)

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Omitted variable bias

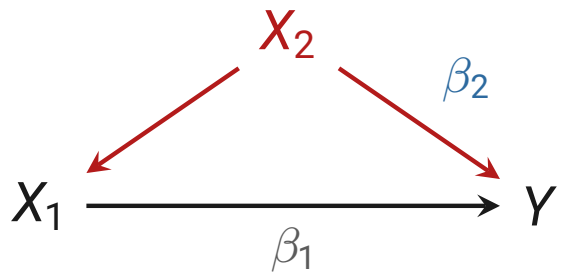
Three routes to a zero conditional mean

Wrapping up

An assumption we cannot check from the data

- OLS is unbiased if Assumptions 1–4 hold.
- We can check A3 in the sample. A1 and A2 concern the form of the model and how we drew the sample.
- A4 is a claim about ε, and we never observe ε.
- The most common failure of A4: one of the terms in ε is a variable that affects Y and is correlated with X.

Suppose ε equals $\beta_2 X_2$ plus other terms, where X_2 is a second variable that affects Y and is correlated with X_1. What does $\hat{\beta}_1$ estimate?

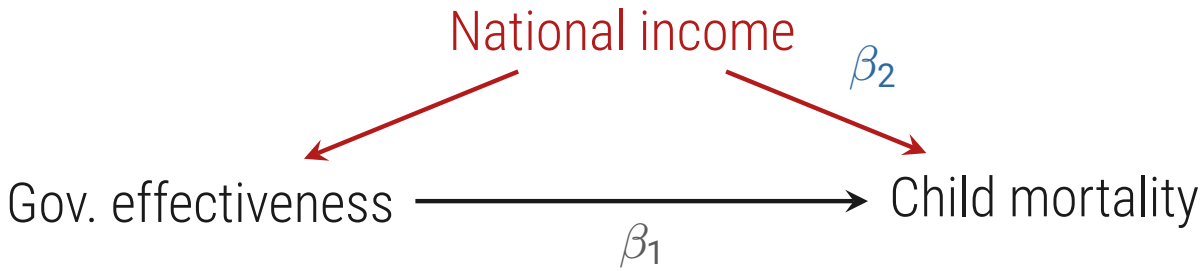


On average, $\hat{\beta}_1$ equals β_1 plus a second term: how much X_2 moves with X_1, times the effect of X_2 on Y.

An omitted variable in the country data

- We regress under-5 mortality on government effectiveness in 46 sub-Saharan African countries. $\hat{\beta}_1 < 0$: countries with more effective governments have lower child mortality.
- Richer countries have more effective governments. Richer countries also have lower child mortality.
- National income is not in the regression, so national income is part of ε.

Does A4 hold? Is the average of ε the same in countries with effective and ineffective governments?



Why not include X_2?

We often cannot add X_2 to the regression:

- We are reading someone else's analysis.
- X_2 is not in our data.
- We do not know which variables to add.

We may not be able to remove the bias. We can still work out its size and its sign.

The long and the short regression

Suppose the data were produced by the **long regression**

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X_1, X_2] = 0.$$

A4 holds for the long regression: for every combination of X_1 and X_2, the average of ε is 0. This ε is the ε of the earlier frames with $\beta_2 X_2$ taken out.

We do not observe X_2, so we fit the **short regression**, Y on X_1 alone:

$$Y = \beta_0^* + \beta_1^* X_1 + v.$$

Call its estimated slope $\hat{\beta}_1^*$.

$\hat{\beta}_1^*$ is the $\hat{\beta}_1$ from the country example.

We want β_1. The short regression gives us $\hat{\beta}_1^*$.

$$\text{Bias} = \mathbb{E}[\hat{\beta}_1^*] - \beta_1.$$

While we cannot eliminate this bias, we can do a bit of math to understand how it affects our estimates.

Step 1: regress the omitted variable on the included one

We start by estimating the relationship between X_1 (the variable we do observe) and X_2 (the omitted variable):

The **auxiliary regression**:

$$X_2 = \delta_0 + \delta_1 X_1 + w, \qquad \mathbb{E}[w \mid X_1] = 0.$$

- δ_1 is the population slope from regressing X_2 on X_1: the change in the average of X_2 when X_1 rises by one unit. If we observed X_2, we could estimate δ_1.
- $\delta_1 = \mathrm{Cov}(X_1, X_2)/\mathbb{V}[X_1]$, so δ_1 has the sign of the correlation between X_1 and X_2.

Steps 2–4: substitute and group

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

The long regression.

$$= \beta_0 + \beta_1 X_1 + \beta_2 (\delta_0 + \delta_1 X_1 + w) + \varepsilon$$

Step 2. Replace X_2 with the auxiliary regression.

Now that we've written X_2 in terms of X_1, the long regression includes only the variables Y and X_1.

$$= \beta_0 + \beta_1 X_1 + \beta_2 \delta_0 + \beta_2 \delta_1 X_1 + \beta_2 w + \varepsilon$$

Step 3. Multiply β_2 through the parentheses.

$$= \underbrace{(\beta_0 + \beta_2 \delta_0)}_{\beta_0^*} + \underbrace{(\beta_1 + \beta_2 \delta_1)}_{\beta_1^*} X_1 + \underbrace{(\beta_2 w + \varepsilon)}_{v}$$

Step 4. Group the constants, the terms in X_1 and the error terms.

Notice that this has the form of our short regression.

Steps 5–6: the bias

Step 5. The error $\beta_2 w + \varepsilon$ has mean 0 at every value of X_1, because w and ε both do. So the last line is the short regression, $Y = \beta_0^* + \beta_1^* X_1 + v$. Its slope is

$$\beta_1^* = \beta_1 + \beta_2 \, \delta_1.$$

- β_2: the slope on X_2 in the long regression (omitted variable $\rightarrow$ outcome).
- δ_1: the population slope from regressing X_2 on X_1 (included variable $\rightarrow$ omitted variable).

STEP 6. OMITTED VARIABLE BIAS

$$\text{Bias} = \mathbb{E}[\hat{\beta}_1^*] - \beta_1 = \beta_1^* - \beta_1 = \beta_2 \, \delta_1.$$

Step 5 shows that v has mean 0 at every value of X_1. So $\mathbb{E}[\hat{\beta}_1^*] = \beta_1^*$.

When does omitting X_2 matter?

$$\text{Bias} = \mathbb{E}[\hat{\beta}_1^*] - \beta_1 = \beta_2 \, \delta_1$$

Omitting X_2 biases $\hat{\beta}_1^*$ only if both hold:

1. $\beta_2 \neq 0$: X_2 is related to Y, holding X_1 fixed;
2. $\delta_1 \neq 0$: X_2 is correlated with X_1.

These two conditions identify variables whose omission biases $\hat{\beta}_1^*$. The main example is a **confounder**: a variable that affects both X_1 and Y.

The sign of the bias in $\hat{\beta}_1^*$

Bias = $\beta_2 \, \delta_1$. Since $\delta_1 = \text{Cov}(X_1, X_2)/\mathbb{V}[X_1]$ and $\mathbb{V}[X_1] > 0$, δ_1 has the same sign as $\text{Cov}(X_1, X_2)$. So the sign of the bias depends on the sign of β_2 and the sign of $\text{Cov}(X_1, X_2)$.

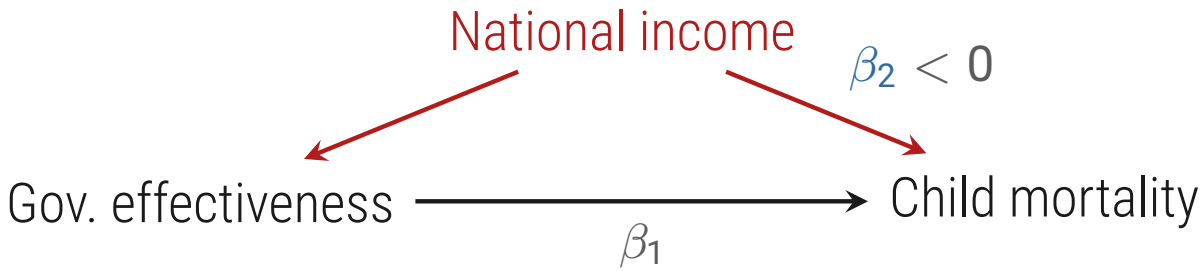
	$\text{Cov}(X_1, X_2) > 0$	$\text{Cov}(X_1, X_2) < 0$
$\beta_2 > 0$	positive bias	negative bias
$\beta_2 < 0$	negative bias	positive bias

Positive bias: $\mathbb{E}[\hat{\beta}_1^*] > \beta_1$. Negative bias: $\mathbb{E}[\hat{\beta}_1^*] < \beta_1$.

Back to the country data

Omitted variable: national income.

- $\beta_2 < 0$: richer countries have lower child mortality, holding government effectiveness fixed.
- $\delta_1 > 0$: richer countries have more effective governments.



Bias = $\beta_2\,\delta_1 < 0$. So $\beta_1^* < \beta_1$: on average, $\hat{\beta}_1^*$ is below β_1.

Our estimate is $\hat{\beta}_1^* < 0$. Suppose β_1^* is also negative. Then $\beta_1 > \beta_1^*$. So β_1 is negative but closer to zero than β_1^*, exactly zero, or positive. Without data on national income, we cannot tell which.

Roadmap

Notation and where we are

The Gauss–Markov assumptions

Omitted variable bias

Three routes to a zero conditional mean

Wrapping up

The problem

How can we make $\mathbb{E}[\varepsilon \mid X] = 0$ plausible?

1. **Controls.** Measure every other influence on Y that is correlated with X, and include it.
2. **Research design.** Find cases where X varies for reasons unrelated to the other influences on Y.
3. **Random assignment.** Set X ourselves, at random.

The coefficient of interest and the controls

$$Y_i = \beta_0 + \beta_1 X_i + \gamma W_i + \varepsilon_i$$

- β_1: the coefficient we care about.
- W_i: one or more control variables. γ: their coefficients.

If $\mathbb{E}[\varepsilon \mid X, W] = 0$, OLS is unbiased for every coefficient.

Often we want an unbiased estimate of β_1 only. The controls are there to make Assumption 4 plausible for β_1.

Route 1: controls

Does campaign contact raise turnout? 2024 American National Election Study: "Has anyone from one of the political parties called you up or come around and talked to you about the 2024 campaign . . . ?"

$$\texttt{voted}_i = \beta_0 + \beta_1 \texttt{contacted}_i + \gamma W_i + \varepsilon_i$$

Campaigns contact people they expect to vote. Variables that might be related to turnout and correlated with contact:

- interest in politics; turnout in past elections
- age; education
- party registration

The problem: how do we know we have controlled for every confounder?

Route 2: research design

Media markets that cross state lines, 2012 presidential election (Enos and Fowler 2018):

$$\texttt{voted}_i = \beta_0 + \beta_1 \texttt{battleground}_i + \varepsilon_i, \qquad \text{within one media market}$$

- Presidential campaigns concentrate on battleground states. Some media markets straddle the border between a battleground state and a non-battleground state.
- Voters on the two sides of the border watch the same local TV. Campaign contact differs because of the Electoral College, not because of the voters.
- The authors check: the two sides had similar demographics and similar turnout in 2010, an election with no presidential campaign.

The problem: how sure are we that the two sides differ only in campaign contact?

Route 3: random assignment

A field experiment in New Haven, 1998 (Gerber and Green 2000):

$$\texttt{voted}_i = \beta_0 + \beta_1 \texttt{canvass}_i + \varepsilon_i$$

- Registered voters were assigned at random to a door-to-door canvassing visit or to no visit: every voter had the same probability of assignment.
- Then $\mathbb{E}[\varepsilon \mid \texttt{canvass}] = 0$: averaged over every possible random assignment, the two groups have the same average ε. Gerber and Green assigned voters at random only once. Like one random sample, one random assignment can produce groups that differ by chance on parts of ε. Such chance differences are sampling error, not bias. The SE measures their typical size.

The problem: which interesting variables can we assign at random?

Roadmap

Notation and where we are

The Gauss–Markov assumptions

Omitted variable bias

Three routes to a zero conditional mean

Wrapping up

The assumptions in one table

Assumption	Says	Needed for	How it fails
A1 Linearity	$Y = \beta_0 + \beta_1 X + \varepsilon$	unbiasedness	a curved relationship
A2 Random sampling	iid (X_i, Y_i)	unbiasedness; the usual SE	selection on Y (bias); dependent units (SE)
A3 Variation in X	$\sum_i (X_i - \bar{X})^2 > 0$	the line exists	constant X; exact collinearity
A4 Zero conditional mean	$\mathbb{E}[\varepsilon \mid X = x] = 0$	unbiasedness	an omitted variable that affects Y and is correlated with X; Y affects X
A5 Homoskedasticity	$\mathbb{V}[\varepsilon \mid X = x] = \sigma_\varepsilon^2$	BLUE; the usual SE	spread changes with X
A6 Normality	$\varepsilon \mid X \sim N(0, \sigma_\varepsilon^2)$	t and CIs when n is small	skewed or heavy-tailed errors (a problem only when n is small)

Omitted variable bias: $\mathbb{E}[\hat{\beta}_1^*] = \beta_1 + \beta_2 \delta_1$.

Further reading

- Gailmard, *Statistical Modeling and Inference for Social Science*, section 7.5.
- Aronow and Miller, *Foundations of Agnostic Statistics*, sections 4.1–4.2.
- `handout_multiple_regression.pdf`: multiple regression in matrix form.
- `handout_ovb.pdf`: quantifying omitted variable bias.

These slides draw on teaching materials by Matt Blackwell, Amitabh Chandra, Daniel de Kadt, Tom Pepinsky and Teppei Yamamoto. Data: World Bank, World Development Indicators and Worldwide Governance Indicators; American National Election Studies 2024.